

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T		
Standard	+	Omni	Power	Omni	Power	Master

Section 14

NTDAC

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1. INTRODUCTION

This chapter describes how to test the NTDAC hardware.

It includes testing of the BDAS, the reconstructor XP1000 6/500 and the RF pulse power amplifier.

In case the synergy option must be installed on this system, it is advised to do this first. After the installation of the synergy hardware it can be tested and calibrated together with the NTDAC at the same time. The description of the synergy hardware installation is present in the delivery of every synergy coil. After installation of the synergy hardware, return to this chapter.

First the NTDAC will be powered up.

2. WARNINGS

1. When using the software tool Receiver Chain Analysis (RCA) dangerous situations can occur, when the parameter 'SAFETY CHECKING' is DISABLED. Only authorized personnel is allowed to disable the 'SAFETY CHECKING'.
2. Hazardous voltages can exist at the output of the power amplifier. Therefore it is advised to disconnect the RF Input cable (FRX1) and the RF enable cable (FRX4) of the amplifier before disconnecting the RF Output cable (FRX2).

3. INFORMATION BEFORE STARTING

3.1. PRECONDITIONS

1. Make sure that the system is sufficient warmed-up for at least 20 minutes, with all cabinets closed, before performing any checks or adjustments.
2. Mains supply voltage to NTDAC within specification.

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4. FIRST TIME POWER ON

4.1. WINDOWS NT WORKSTATION 4.0 LICENSE

Because the operating system installed on the reconstructor is Windows NT workstation 4.0, a unique Microsoft license number is supplied with every reconstructor.

The Microsoft license number can be found on the booklet that is placed in the NTDAC.

4.2. RECONSTRUCTOR SOFTWARE INSTALLATION AND CONFIGURATION

From release 7.1 onwards the system disk of the reconstructor will contain the complete software already loaded. This means that only network configuration has to be done.

The procedures is described in 4.5.

We strongly recommend to study each new Software Installation Manual. Besides the installation procedures for basic software and options, it informs you about

- New features/improvements/options
- Known problems and workarounds
- Problem reporting

4.3. CHECK PRE-INSTALLED SOFTWARE RECONSTRUCTOR

1. Switch on the LCD which is located in the drawer in the NTDAC.
2. From the AlphaBIOS boot screen press the <F2> key to enter the AlphaBIOS setup
3. Select: **Display System Configuration**
4. Check in the **System board configuration** screen that the Alphabios version is **5.70**
5. Select: <ESC> a number of times to exit the Alphabios setup menu
6. Switch on the reconstructor computer and check if it will boot automatically.
7. Check that the reconstructor starts Windows NT

4.4. RECONSTRUCTOR POWER ON SEQUENCE

After switching on the reconstructor, a power-up sequence starts. During this power-up sequence, Power Up Self Tests (POST) are performed to initialise the hardware and test its ability to perform basic tasks.

After successful completion of the self-tests, Windows NT will load automatically.

Windows NT is the only operating system which is installed on the reconstructor XP1000 6/500.

NOTE

It is not necessary to login on the reconstructor to perform TSW tests or application measurements.

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4.5. RECONSTRUCTOR CONFIGURATION

To complete the network configuration, the reconstructor is configured with default TCP/IP address settings.

4.5.1. RECONSTRUCTOR CONFIGURATION ON RECONSTRUCTOR

Configuring the default host computer TCP/IP address.

1. Switch on the LCD in the drawer of the NTDAC.
2. Login on the reconstructor with user MRSYSTEM, password = manager.
3. Select: **Start**
4. Select: **MR configuration**
5. Select: **Host configuration**
6. Enter the TCP/IP address: **192.192.192.201**

Configuring the default reconstructor TCP/IP address:

1. Select: **Start**
2. Select: **MR configuration**
3. Select: **Network**
4. Select: **Identification tab**
5. Click on button : **change**
6. Enter the host name : **RECONSTRUCTOR** (if not already filled in)
7. Click on : **OK**
8. Select: **Protocol tab**
9. Click on: **properties**
10. Enter the TCP/IP address: **192.192.192.210** (in the IP address box)
11. Click on: **OK**
12. Click on: **OK**
13. Press <Control>+ALT+<Delete>.
14. Select: Lock Workstation.
15. Switch off the LCD in the drawer of the NTDAC.
16. Slide the drawer back in the NTDAC.

4.5.2. RECONSTRUCTOR CONFIGURATION ON HOST

1. Login on host computer with user **system** + <password>.
2. Select: Configure Hardware Options
3. Select: Reconstructor Management menu
4. Select: Modify Network Configuration Reconstructor
5. Press <Return> after:
Enter the host-name of the Reconstructor [RECONSTRUCTOR]:
6. Enter internet address for Reconstructor:
Enter internet address for RECONSTRUCTOR []: **192.192.192.210**
7. Press <Return> after:
Press [RETURN] for Reconstructor Management menu
8. Select: Exit to configure Hardware Options menu
9. Select: Exit to Gyroscan Manager menu
10. Select: Logout and confirm by clicking **yes**.

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4.6. CHECKING NETWORK CONFIGURATION

At this point, all basic network units are assigned with a default TCP/IP address and host name.

Checking the defaults:

1. Login on the host with user system + <password>.
2. Select: **Exit** to DCL
3. Enter: **ucx show host**
4. The following list with default TCP/IP addresses should be displayed.

TCP/IP address	Host name, Alias	Remark
127.0.0.1	LOCALHOST, localhost	Fixed TCP/IP address
192.192.192.212	SWITCH, switch	
192.192.192.201	NTSCAN, ntscan, OC, oc	
192.192.192.211	JETDIRECT, jetdirect	Print server
192.192.192.210	RECONSTRUCTOR, reconstructor	

Table 1: Default TCP/IP addresses and host names

5. Enter: **Logout** and confirm by clicking **yes**.

4.7. BDAS POWER ON SEQUENCE

After switching on the BDAS a selftest is performed. This will take approximately 1 minute. After the selftest has finished, two LEDs on the front of the BDAS Control Processor (BCP) will be on. (one red led and one green led)

After downloading software to the BDAS, starting Gyrorestart, or starting TBDAS (Test software) the red LED will extinguish.

4.8. RF PULSE POWER AMPLIFIER, POWER ON SEQUENCE

Hazardous voltages can exist at the output of the power amplifier. Therefore; ALWAYS disconnect the RF Input cable (FRX1) and the RF Gating cable (FRX4) of the amplifier before disconnecting the RF Output cable (FRX2).

MR5002 :

1. Switch the front panel circuit breaker from the '0' position to the '1' position. The front panel DC-on led will light for approximately 8 seconds and it will extinguish.
2. Next the front panel 'NORMAL' LED will light, and within approximately 5 seconds, the front panel 'NORMAL' LED will blink once and then light again.

S21, S22, S23 and S24 :

1. Switch the front panel breaker on. The RF pulse power amplifier will power on to the STANDBY state.
It will take the S21 and S22 amplifier approximately 6 minutes to go to the STANDBY state. The S23 and S24 amplifier take about 3 minutes to go to the STANDBY state. The fault LED must not be lit.

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4.9. GRADIENT INTERFACE POWER ON

Before applying power to the gradient amplifier cabinet check that the system interconnections and the grounding straps from the side panels and doors are connected properly.

For Intera Standard and Omni:

1. Make sure that the 15 kW/160V power supply unit (DP) is switched OFF
2. Switch on the MDU.
3. Switch on the "MAINS and AUX" switches of the gradient amplifier cabinet. Both switches can be found at the bottom front panel.
4. The green "power"led's (DI) of the Controller and WFG/ECC boards must be on

For Intera Power and Master:

1. Make sure that all 20kW/420V power supplies (DP1 and DP2) are switched OFF.
2. Switch on the MDU and MDU extended.
3. Switch on the "MAINS and AUX" switches of the gradient amplifier cabinet, which contains the DFEC3 Eddy current rack (DI). The switches can be found on the PDU unit.
4. The green "power"led's (DI) of the Controller and WFG/ECC boards must be on

4.10. FRONT-END POWER ON SEQUENCE

To supply the front-end with power, the front end power supply unit must be switched on.

4.10.1. FEPSU

The power switch for this unit is reachable when the lower cover of the system filter box is removed. A green LED indicates that the FEPSU is on and that all supplies are present and within specification. The 300V is supplied by an additional unit and is not controlled by this LED.

1. Switch the FEPSU on if not already done.
2. Check that the LED is on, indicating that all supplies are present and within specification.

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5. NTDAC TEST

5.1. NETWORK CONNECTION HOST <-> RECONSTRUCTOR

1. Login on the host with user system.
2. Select: Exit to DCL
3. Enter: **ucx ping reconstructor**
4. The system should reply with:
%UCX-I-LOOPACT, RECONSTRUCTOR is alive

5.2. BDAS AUTOMATIC TEST PROCEDURE

1. Login : TSW
2. Select: TBDAS
3. Select: Run functional automatic.
The total automatic test takes about one minute.
4. No errors are allowed.
5. Enter : Quit (to exit the TBDAS test).

5.3. RECONSTRUCTOR AUTOMATIC TEST PROCEDURE.

In this test, also the network connection from reconstructor to host, via the ethernet switch is tested.

1. Login : TSW
2. Select: TREC
3. Enter : Run functional automatic.
This test will take about 2 minutes.
4. No errors are allowed.
5. Enter : Quit (to exit the TREC test).

6. SYSTEM TUNING TOOLS

The System Tests and Tools (STT) are divided into different groups. Refer to Figure 2 to find the menu entries.

The entries "Enter service mode" and "System tuning" are protected with a password.
When the service dongle is used and the string is entered, the system will not ask for these passwords.

7. DATABASE PARAMETERS

1. Login : GYROSCAN.
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select: ENTER SERVICE MODE.
5. Select: SYSTEM TUNING.
6. Select: SYSTEM TUNING TOOLS.
7. Select: DATABASE.

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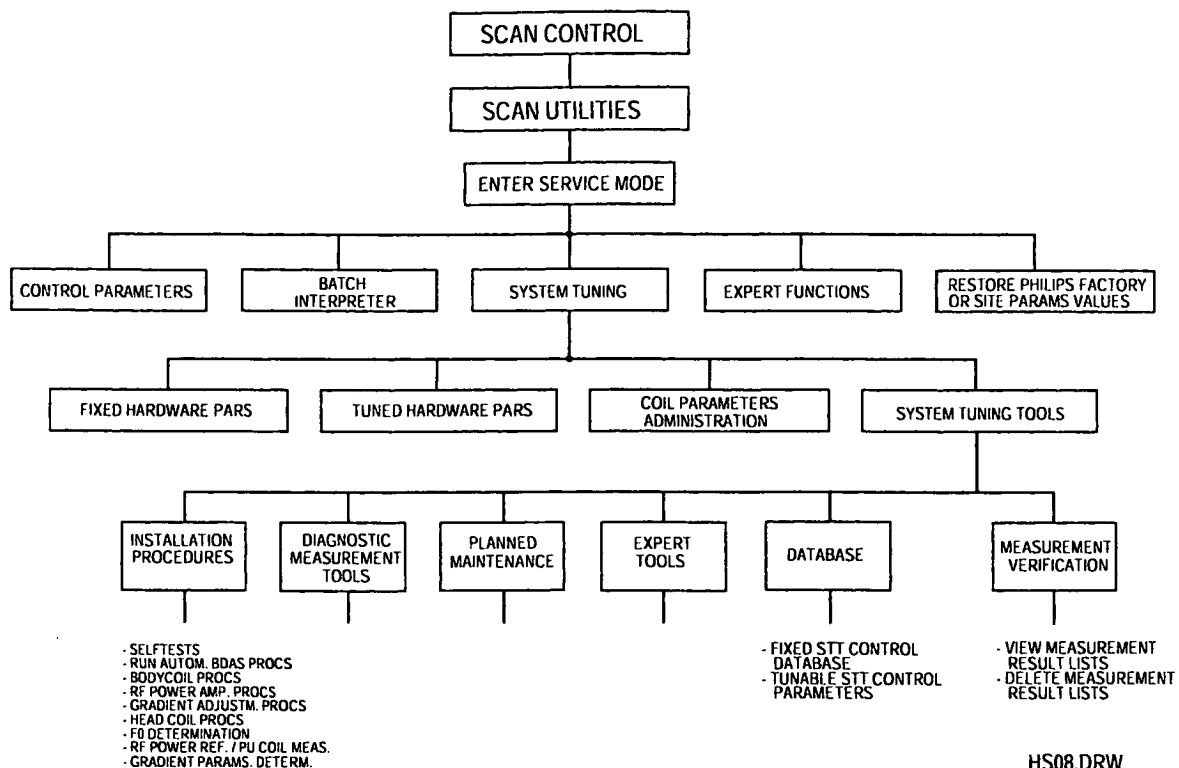
8. Select: TUNABLE STT CONTROL PARAMETERS
9. Change/check:
 - KWA: Cable atten [dB] Attenuation of calibrated measuring cable. Value on label
 - KWA: 50E-load atten. [dB] Attenuation of 50E dummy load. Value is written on the load (if available)
 - KWA: Coupler attenuator [dB] Attenuation of the Fwd power connector FRX5 on the Ehrhorn Power amplifier. Coupler value is written on a sticker close to the connector (Not applicable for Intera 0.5T systems)
10. Select: STORE to save the changes.
11. Follow the information on the screen
12. Select: CANCEL to quit

8. HARDWARE PARAMETER LIST

During installation procedures some hardware parameters will be determined and stored in the hardware parameter file. To know the default values, a printout can be made.

1. Login : GYROSCAN (if not already done)
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select: ENTER SERVICE MODE.
5. Select: SYSTEM TUNING.
6. Select: TUNED HARDWARE PARAMETERS
7. Select: LIST A File is created in directory GYRO\$TEMP
8. Select: PRINT The file is printed.

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HS08.DRW

Figure 1 System tuning tools menu structure

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9. BDAS ADJUSTMENTS

NOTE

If you need to install the synergy hardware during the installation of the system, this should be done first. The synergy hardware will be tested and calibrated during the following test. The procedure for synergy installation is present in the delivery of every synergy coil.

Before starting this procedure, make sure that no surface coils are present in the magnet bore and that the system is switched ON at least 20 minutes.

1. Login : GYROSCAN
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select: ENTER SERVICE MODE.
5. Select: SYSTEM TUNING.
6. Select: SYSTEM TUNING TOOLS
7. Select: INSTALLATION PROCEDURES.
8. Select: RUN AUTOMATIC BDAS PROCEDURES.

Answer the question "run all subsequent tools successively" with Yes.

The transceiver characteristics are measured and transmit/receive attenuator are calibrated.
By running this test, the following checks are done / calibrations are performed:
The measurement will take about 5 minutes without Synco Receivers.

In case the measured values are not within specifications, a message will appear on the screen.

The next items will be tested:

- Transceiver flatness test
- Transceiver level test
- Transceiver noise test
- Transceiver linearity test
- Transceiver spurs test
- Attenuator and Gain calibration

In case the synergy option is available and installed at the system, extra tests for the Synco receivers are performed:

- Synco Receiver flatness test [left]
- Synco Receiver gain test [left]
- Synco Receiver noise test [left]
- Synco Receiver spurs test [left]
- Synco Receiver flatness test [right]
- Synco Receiver gain test [right]
- Synco Receiver noise test [right]
- Synco Receiver spurs test [right]

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10. RF PULSE POWER AMPLIFIER ADJUSTMENT

10.1. INTRODUCTION

The Power amplifier must be adjusted for correct output level and correct functioning of the PMU (Power Monitoring Unit). The instructions for the measurements will be presented on the screen. For Intera 1.5T systems Power and Master with an S21 amplifier the test is also used to determine the optimal cable length for the Transmit cable. The S23 RF amplifiers do not need a 1/4 lambda cable.

CAUTION

Hazardous voltages can exist at the output of the power amplifier. Therefore; ALWAYS disconnect the RF Input cable (FRX1) and the RF Gating cable (FRX4) of the amplifier before disconnecting the RF Output cable (FRX2).

10.2. MAX KW ADJUSTMENT AND PMU TEST (ALL SYSTEMS EXCEPT INTERA 1.5T POWER AND MASTER)

For Max kW adjustment Intera 1.5T Power and Master systems see next paragraph.

1. Login : GYROSCAN.
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES
4. Select: ENTER SERVICE MODE
5. Select: SYSTEM TUNING.
6. Select: SYSTEM TUNING TOOLS.
7. Select: INSTALLATION PROCEDURES.
8. Select: RF POWER AMPLIFIER PROCEDURES.
9. Select: 1. MAX. KW ADJUSTMENT (1H RF-AMP)

This procedure is used to measure automatically the RF drive scale which results in 5 kW output. Check the settings for cable attenuation and coupler attenuation. Follow the instructions on the screen.

10. Select: 1. MAX. KW ADJUSTMENT (1H RF-AMP)

After the test has finished:

11. Select: 2. PMU TRIP TEST.

This procedure checks whether the internal PMU switches the RF pulse power amplifier off within 30 seconds in case the average output power is 10 % higher than the PMU setting. It also checks whether the amplifier remains on in case the average output power is 10 % lower than the PMU setting.

12. Follow the instructions on the screen.

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10.3. MAX KW ADJUSTMENT AND PMU TEST (ONLY FOR INTERA 1.5T POWER AND MASTER)

This test calibrates the setting of the BDAS for 5kW output on the Power amplifier. In case of an S21 amplifier this test is also used to determine the optimal cable length of the transmit cable. Due to the use of 25 kW for these systems, the cable length adaptation is necessary. In case of an S23 RF amplifier, the cable length measurement is not necessary.

In case of an S21 amplifier; in the program, the operator is asked to connect an additional cable of 1/4 lambda length in the transmit cable. One side of this 1/4 lambda cable should be connected to the System filter box ZAX1 and the other side should be connected to the transmit cable labeled ZAX1 using an adapter. In this way the 1/4 lambda cable is in series with the transmit cable.

WARNING

Disconnect the RF Input cable (FRX1) and the RF enable cable (FRX4) of the amplifier before disconnecting the RF transmit cable.

The program measures one time without the 1/4 lambda cable and one time with this cable. After both measurements are performed, the program determines whether the cable should be removed or should be present. The Max kW adjustment is performed in the same measurement.

A shrink sleeve is delivered with the system to insulate the connector of the 1/4 lambda cable. This sleeve must be used to prevent spikes due to the connector touching the system filter box. Use a heatgun (not too hot) to shrink the sleeve over the connector of the 1/4 lambda cable if the program has determined that the cable should be present in the system.

1. Login : GYROSCAN.
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select: ENTER SERVICE MODE.
5. Select: SYSTEM TUNING.
6. Select: SYSTEM TUNING.
7. Select: SYSTEM TUNING TOOLS.
8. Select: INSTALLATION PROCEDURES.
9. Select: RF POWER AMPLIFIER PROCEDURES.
10. Select: 1. MAX. KW ADJUSTMENT (1H RF-AMP)
11. Start the program with <Proceed
12. Fill in the values for cable attenuation and directional coupler output.
13. **Enter YES after the question: Perform Cable length tuning?
14. Follow the instructions on the screen.
15. Select: PROCEED to start the measurement.

After the test has finished:

16. Select: 2. PMU TRIP TEST.

This procedure checks whether the internal PMU switches the RF pulse power amplifier off within 30 seconds in case the average output power is 10 % higher than the PMU setting. It also checks whether the amplifier remains on in case the average output power is 10 % lower than the PMU setting.

17. Follow the instructions on the screen.

** In case of an S23 RF amplifier the cable length adjustment is not necessary. Therefore this question will not appear when an S23 amplifier is used.

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11. TEST SCAN

For a quick check of the system it is possible to perform a test scan. A phantom is not needed and no RF power is sent to the body coil. The magnet may be "ON" or "OFF" field.

11.1. PROCEDURE

1. Disconnect the RF-input (FRX1) to the RF pulse power amplifier.
2. Login : GYROSCAN.
 Select: PATIENT.
 Select: NEW EXAM.
 Select: DUMMY EXAMINATION
 Select: SCAN CONTROL
 Select: SCAN UTILITIES.
 Select: ENTER SERVICE MODE.
 Select: CONTROL PARAMETERS.
 Select: PREPARATION CONTROL PARAMETERS.

Change the preparation control parameters as follows:
 Log the original values for restoring.

Coil identification	On
Tuning and Matching	On
DC fixed correction	On
Actual scan	On

The other parameters must be set to "off"

3. Select: PROCEED to store the parameter file.

Perform Body coil performance measurement B10.

4. Select: < ICON person with slices >.
5. Select: ANATOMY.
6. Select: PHANTOM STUDIES
7. Select: BODYT.
8. Select: B10 : MS,SE,30
9. Select: START SCAN.

No errors are allowed. If you find an error, then check the log files and repair the error.

10. Reconnect the RF-input (FRX1) to the RF pulse power amplifier
11. Change the preparation control parameters to the original values.